

Object Oriented Programming Lab

Lab 14

Instructions

Work on this lab individually. You can use your books, notes, handouts etc but you are not allowed to borrow anything from your peer student.

Write down the lab number, task number, your name and roll number at the start of each task code using comments.

For Example

```

/*****
Lab:    xx
Task:   yy
Developer:
        Roll #: XXXYYYZ000
        Name:  XYZ
*****/

```

What you have to do

Program the following tasks in C++, compile and execute them.

Task

An organization has different types of employees. Every employee has an *Employee ID*, *Salary*, and *Rank*. Their calculation varies for different types of employees, therefore these functions shall be declared as pure virtual in the base class.

Create base class **Employee** with following attribute and functions:

- string empID
- Constructor printing "Employee constructor"
- void showID()
- virtual int getSalary() = 0;
- virtual string getRank() = 0;
- virtual void status()

Create class **Developer : public Employee**

- Constructor printing "Developer constructor"
- Override:
 - getSalary()
 - getRank()
 - status()

Create class **Manager : public Employee**

- Constructor printing "Manager constructor"
- Override:
 - getSalary()
 - getRank()
 - status()

Create class **TechLead : public Developer, public Manager**

- Constructor printing "TechLead constructor"

In **main()**

1. Ambiguity Demonstration

```
TechLead tl;  
tl.showID();    // should cause ambiguity
```

2. Fix Ambiguity Using Virtual Inheritance

Modify the inheritance as follows:

```
class Developer : virtual public Employee {};  
class Manager   : virtual public Employee {};
```

Now the following code should work correctly:

```
TechLead tl;  
tl.showID();    // ambiguity resolved
```

3. Polymorphism Using Base Class Pointer

Add the following in **main()**:

```
Employee* e = new TechLead();  
e->status();  
cout << "Salary: " << e->getSalary() << endl;  
cout << "Rank: " << e->getRank() << endl;
```

😊😊😊 **BEST OF LUCK** 😊😊😊
